

StockIQ

Personalized & Explainable AI Stock Decision-Support System

Software Requirements Specification

Version 1.0 • October 2026 • Status: Baseline for MVP development
Standards: IEEE 29148-aligned

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Educational decision-support software. Not investment advice. Not a SEBI-registered advisory service.

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Document at a glance

• **Product:** Web platform that predicts stock direction, explains it with SHAP, personalizes it to the investor, analyses portfolio risk, simulates what-if changes, compares stocks vs Gold vs FD, and answers SEBI questions via RAG.

• **Architecture:** Modular monolith — Streamlit + FastAPI + PostgreSQL/pgvector, all Python, no paid service required for development.

• **Differentiator:** Not “stock will go up” but *predict → explain → check suitability → analyse portfolio → simulate → flag conflicts*.

1. Introduction

1.1 Purpose

This SRS defines the functional, non-functional, data, AI and deployment requirements for **StockIQ**, an AI-powered stock investment decision-support system for Indian retail investors (NSE/BSE). It is the baseline for design, development, testing and acceptance.

1.2 Scope

In scope (MVP)	Out of scope (Post-MVP / Future)
Email/password login • Investor profile (age, occupation, income, horizon, risk, amount, goal) • Next-day signal (BUY/HOLD/SELL) with calibrated confidence for a fixed NSE watchlist (Nifty-50 subset) • SHAP explanation + plain-language text • Personal suitability check • Portfolio sector/risk analysis • What-if simulator • Stock vs Gold vs FD comparison • Daily open/close probability ranges • Backtesting • SEBI RAG chatbot • Dashboard • Basic admin panel	Real-money trading / broker integration • Intraday & F&O; • Crypto / US stocks • Live news-sentiment model (FinBERT) • Push/WhatsApp/SMS alerts • Payments & subscriptions • Mobile app • Multi-language • Tax computation • Social features

1.3 Definitions

Term	Meaning
SHAP	SHapley Additive exPlanations — per-feature contribution to a model output
RAG	Retrieval-Augmented Generation — LLM answers grounded in retrieved SEBI documents
RSI / MACD / SMA / EMA / BB	Technical indicators: Relative Strength Index, Moving Avg Convergence Divergence, Simple/Exponential Moving Average, Bollinger Bands
Beta / MDD / Sharpe	Market sensitivity / Maximum Drawdown / risk-adjusted return
Walk-forward validation	Time-ordered train/test splits that prevent look-ahead leakage
DPDP Act	India’s Digital Personal Data Protection Act, 2023

1.4 Honest Design Assessment (key decisions)

Challenges to the original idea — and how this SRS resolves them

• **“82% confidence” is risky.** Next-day direction models typically score only slightly above chance (~52–58% accuracy). Confidence must be a *calibrated probability* (isotonic/Platt), never presented as accuracy. UI must show backtest hit-rate beside it.

• **Random train/test split causes leakage.** Mandatory: walk-forward validation, features computed using only past data, transaction-cost-aware backtests.

• **“Probability of opening & closing price”** is reinterpreted as: P(next close > today close), P(gap-up), and P10/P50/P90 predicted open/close ranges (quantile regression) — not exact prices.

• **Personalization should be rule-based, not LLM-based.** Age/income/horizon/risk → deterministic risk-capacity score. LLMs are used only for chatbot answers and optional wording polish.

• **SEBI regulation.** Giving personalised buy/sell advice to the public can require SEBI Investment Adviser / Research Analyst registration. StockIQ is positioned as an educational analytical tool with mandatory disclaimers (Appendix A) and “signals”, not “advice”.

• **Data source.** yfinance is unofficial; acceptable for academic use, with an adapter layer so NSE bhavcopy / a licensed API can replace it.

2. Overall Description

2.1 Product Perspective & Problem Statement

Retail investors rely on opaque tips, generic prediction apps and scattered tools. They cannot see *why* a signal is produced, whether it suits their risk capacity, how their portfolio is concentrated, or how it compares with safer assets. StockIQ unifies these steps in one explainable workflow.

2.2 User Classes

User	Goal	Permissions	Main Actions
Guest	Explore the product	Landing page, demo stock page, chatbot (rate-limited)	View demo, register
Investor (registered)	Personalised insights	Own profile, portfolios, simulations, chat history	Create profile, view signals/SHAP, analyse portfolio, run what-if, compare assets, chat
Admin	Operate the system	All users (view), KB documents, model status, logs, flags	Upload SEBI docs, trigger re-index/retrain, view usage & errors, disable users
System (scheduler)	Automation	Internal service account	Daily ETL, signal generation, retraining, cache refresh

2.3 Personas

Persona	Profile	Needs / Pain points	Typical workflow
Riya, 24, IT analyst	Income ₹6L, 5-yr horizon, moderate risk, ₹50k	Wants growth, fears picking blindly; does not understand indicators	Register → profile → view TCS/Infosys signal → read SHAP → check suitability → compare with FD
Mr. Kulkarni, 55, teacher	Income ₹9L, 3-yr horizon, low risk, ₹2L	Capital safety; confused by volatility; unsure about SEBI rules	Profile → sees high-vol stocks flagged → compares FD/Gold → asks chatbot about investor protection
Aman, 30, freelancer	Variable income, holds 4 stocks	Unknown concentration risk	Enter holdings → sector pie → risk score 64 → what-if +₹10k TCS → rebalance

2.4 Operating Environment, Constraints & Assumptions

- **Env:** Modern browser (Chrome/Edge/Firefox); Python 3.11 backend; Linux container hosting.
- **Constraints:** Zero/low budget; two-person team; free-tier rate limits; end-of-day data (not tick data); Indian market hours 09:15–15:30 IST.
- **A1** English only • **A2** Email/password auth • **A3** NSE equities, daily candles • **A4** Watchlist of ~30–50 liquid stocks • **A5** No real trades • **A6** FD rate is an admin-maintained/user-editable assumption • **A7** Gold tracked via GOLDBEES.NS (or gold-INR series) • **A8** Signals refresh once daily after market close.

3. System Architecture & Technology Stack

Style: Modular monolith. One deployable FastAPI service with cleanly separated modules (auth, market, ml, risk, personalization, ai/rag, admin) plus a scheduler. Microservices add operational burden with no MVP benefit; modules can be extracted later.

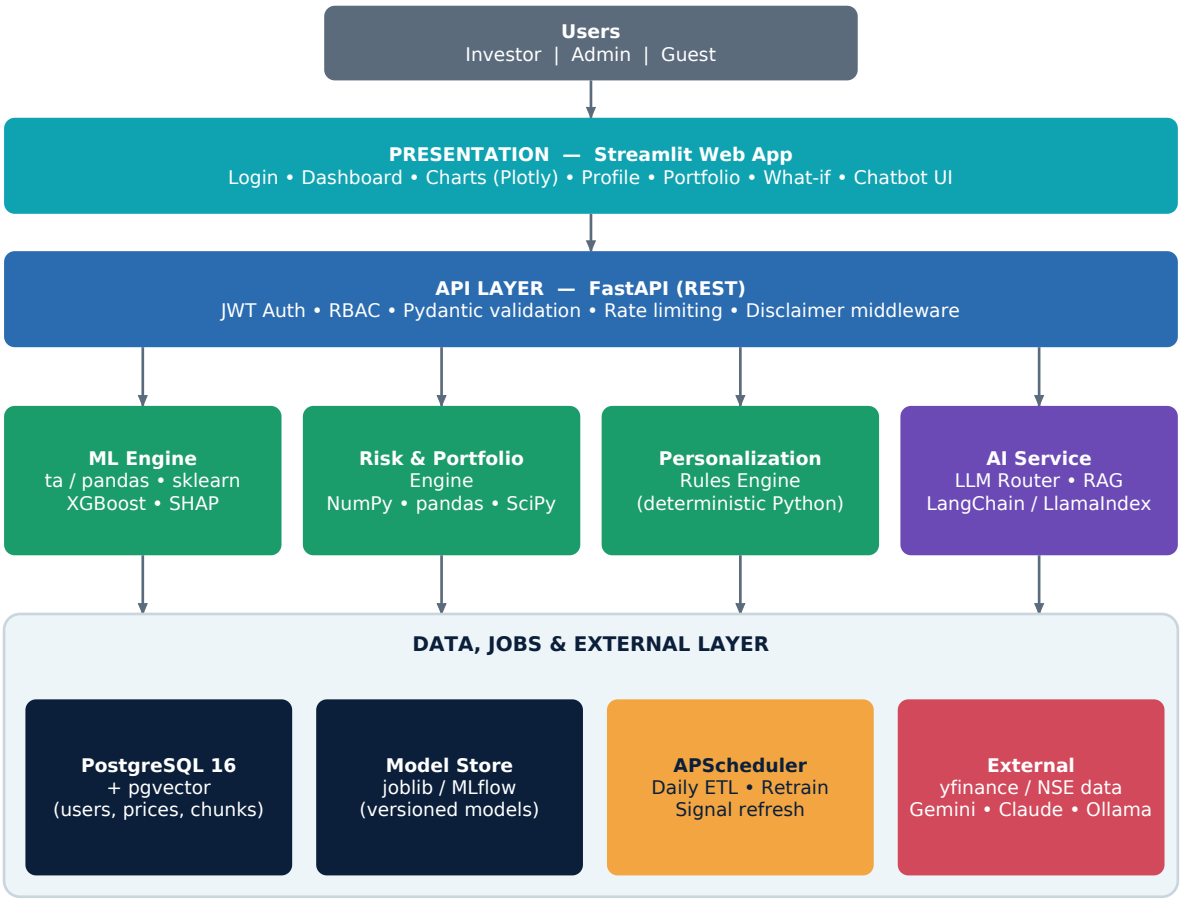


Figure 1 — High-level system architecture

Layer	Technology	Purpose / Why chosen	Cost type
Frontend	Streamlit (MVP); Plotly charts	Fast all-Python dashboards; React/Next.js is the Post-MVP upgrade	Free / OSS
Backend API	FastAPI, Uvicorn, Pydantic v2	Async REST, auto OpenAPI docs, validation	Free / OSS
Auth & security	JWT (python-jose/PyJWT), passlib[bcrypt/argon2], slowapi	Stateless auth, password hashing, rate limiting	Free / OSS
Primary DB	PostgreSQL 16 + SQLAlchemy 2 + Alembic	Relational data, migrations; SQLite allowed for local dev	Free / OSS
Vector store	pgvector (PostgreSQL extension)	Avoids a 2nd database; hybrid with Postgres full-text	Free / OSS
Market data	yfinance (primary), NSE bhavcopy CSV (backup), Alpha Vantage / Twelve Data (optional)	Daily OHLCV, sector, beta; adapter pattern for swapping	Free / Freemium
Data & indicators	pandas, NumPy, SciPy, ta / pandas-ta	RSI, MACD, SMA/EMA, Bollinger, returns, volatility	Free / OSS
ML	scikit-learn, XGBoost, imbalanced-learn, joblib	RF + XGBoost ensemble; calibration; quantile regression	Free / OSS
Explainability	SHAP (TreeExplainer)	Fast exact SHAP for tree models	Free / OSS
Experiment tracking	MLflow (optional) or versioned joblib + metrics JSON	Model versioning, reproducibility	Free / OSS
LLM (chatbot)	Primary: Gemini Flash-class (free tier for dev) • Backup: Claude Haiku-class • Local: Ollama + Llama 3 / Qwen	Cheap, fast, good for grounded Q&A; provider-agnostic router. *Verify current pricing/limits before launch.*	Freemium / Pay-go / Free
Embeddings & rerank	BAAI/bge-small-en-v1.5 (dev) → bge-m3 (MVP) via sentence-transformers; bge-reranker-base	Self-hosted, no API cost, no data leaves server	Free / OSS
RAG framework	LlamaIndex or LangChain (thin use); PyMuPDF / pdfplumber for parsing	Ingestion, chunking, retrieval orchestration	Free / OSS
Sentiment (Post-MVP)	FinBERT (Hugging Face) + RSS/NewsAPI/GDELT	News sentiment as 3rd signal	Free / Freemium
Scheduler / cache	APScheduler; in-process cache (cachetools) → Redis later	Daily ETL, retrain; avoids Celery complexity	Free / OSS

Layer	Technology	Purpose / Why chosen	Cost type
Email (Post-MVP)	Brevo / Resend / SMTP (Gmail app password for dev)	Verification, password reset	Freemium
Containers & CI/CD	Docker, Docker Compose, GitHub Actions	Reproducible builds, tests on every PR	Free / OSS
Hosting	Dev: local Docker • Demo: Render / Railway / Hugging Face Spaces • Prod: AWS Lightsail/EC2 (Mumbai) or Azure/GCP	Simple, India region available for production	Free tier → Pay-go
Monitoring	Sentry (errors), UptimeRobot, Prometheus/Grafana (optional), structlog	Error tracking, health checks, JSON logs	Freemium / OSS
Testing & quality	pytest, pytest-cov, httpx, Locust, RAGAS, Ruff, Black, Bandit, pip-audit	Unit/API/perf tests, RAG evaluation, linting, security scans	Free / OSS

4. Workflow Diagrams

4.1 End-to-End User Journey

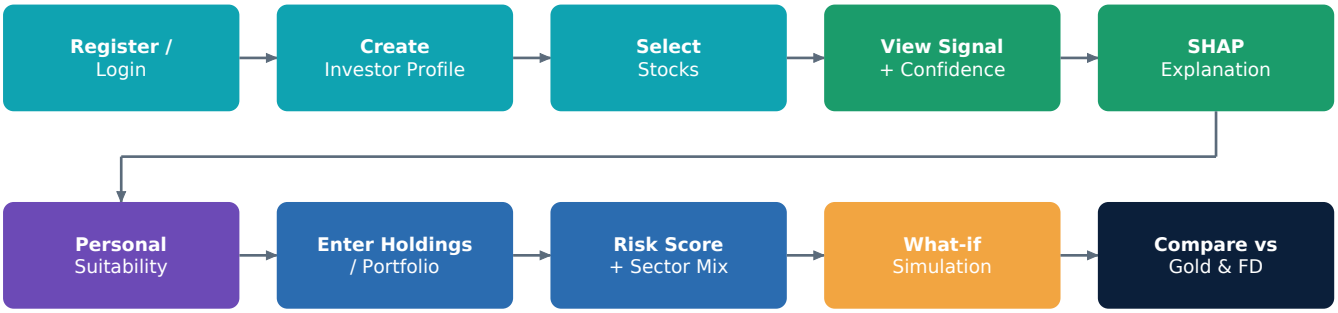


Figure 2 — Entry → Action → Processing → AI → Result → Follow-up (Chatbot Q&A available at every step)

4.2 Prediction, Explanation & Personalization Pipeline

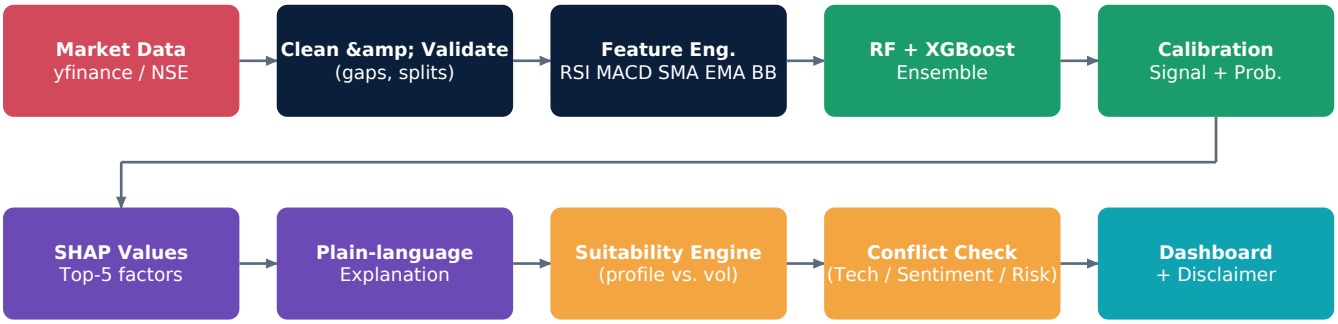


Figure 3 — Daily batch pipeline (after market close) and on-demand explanation

Signal-conflict logic (deterministic):

Technical	Sentiment (Post-MVP)	Risk fit	Final output
BUY	BUY / neutral	Suitable	BUY — high-agreement signal
BUY	SELL	Any	⚠ SIGNAL CONFLICT — “Monitor”, confidence reduced (e.g., 57%)
BUY	Any	High risk for user	BUY with ⚠ Personal Suitability: LOW/MODERATE
Prob. between 45–55%	Any	Any	HOLD — model uncertain

4.3 Portfolio Risk & What-If Workflow

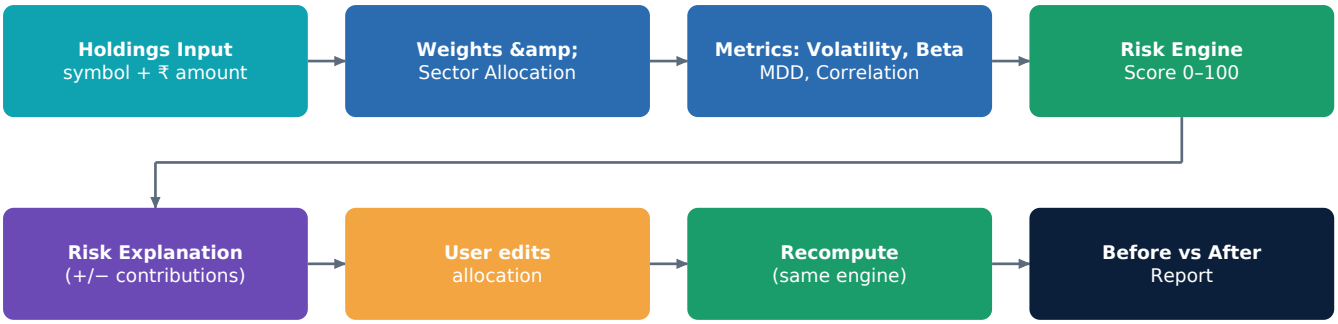


Figure 4 — Portfolio analyzer and what-if simulator (pure deterministic Python — no LLM)

4.4 SEBI RAG Chatbot Workflow

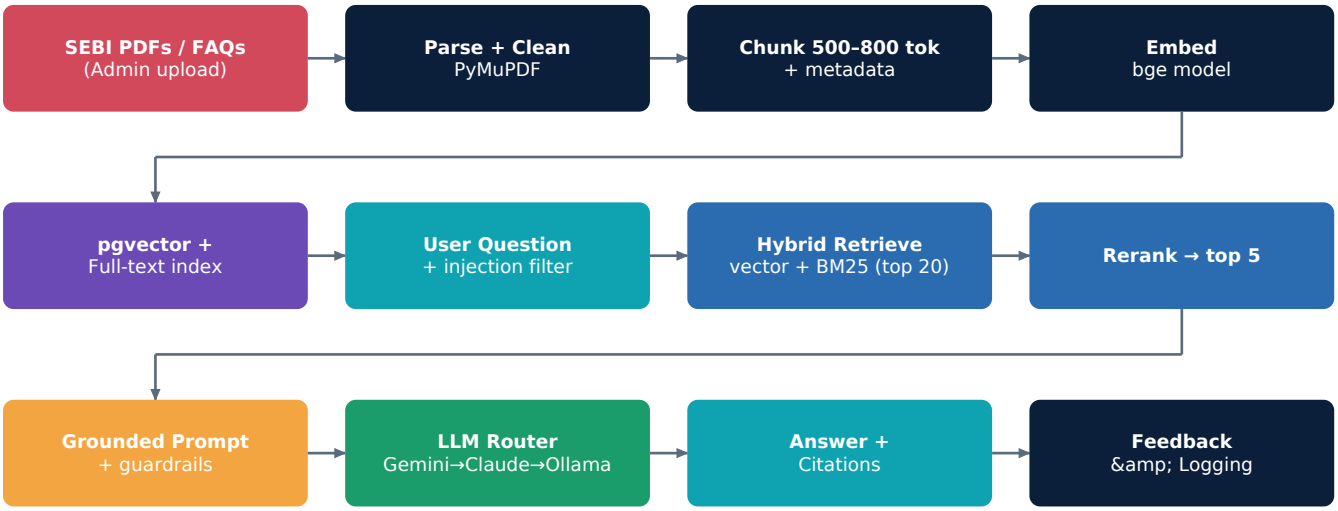


Figure 5 — RAG ingestion (top row) and query-time pipeline

5. Functional Requirements

Priority: **P0** mandatory • **P1** important • **P2** useful • **P3** future. Phase: MVP / Post.

5.1 Authentication & User Management

ID	Requirement	Pri	Phase
FR-AUTH-01	User shall register with email + password (min 8 chars, strength check); duplicate emails rejected.	P0	MVP
FR-AUTH-02	User shall log in and receive a short-lived JWT access token (15–30 min) + refresh token.	P0	MVP
FR-AUTH-03	Passwords stored only as bcrypt/argon2 hashes; account lockout/throttle after 5 failed attempts.	P0	MVP
FR-AUTH-04	Role-based access: Investor vs Admin; admin endpoints return 403 for others.	P0	MVP
FR-AUTH-05	Email verification and forgot-password flow.	P2	Post
FR-AUTH-06	User can export and delete their account and data (DPDP).	P1	MVP

5.2 Investor Profile & Personalization

ID	Requirement	Pri	Phase
FR-PRO-01	Capture age, occupation, annual income, investment amount, horizon, risk tolerance, goal, preferred stocks.	P0	MVP

ID	Requirement	Pri	Phase
FR-PRO-02	Compute Risk Capacity (age, income, horizon) and Risk Willingness (questionnaire) → Risk Profile: Low / Moderate / High.	P0	MVP
FR-PRO-03	For each signal compute Personal Suitability (HIGH/MODERATE/LOW) by comparing stock volatility, beta and drawdown with the user's risk band.	P0	MVP
FR-PRO-04	Suggest an indicative allocation (stocks/Gold/FD) and a ranked stock list matching profile; show reasons.	P1	MVP
FR-PRO-05	Profile is editable; recommendations recompute on change.	P0	MVP

5.3 Market Data

ID	Requirement	Pri	Phase
FR-DATA-01	System shall fetch and store daily OHLCV for the watchlist, with sector and beta metadata.	P0	MVP
FR-DATA-02	Data validation: missing days, zero volume, splits/bonus adjustments, outliers flagged.	P0	MVP
FR-DATA-03	Fallback to secondary data source or last good data if the primary API fails; show "data as of" timestamp.	P1	MVP
FR-DATA-04	Stock search and filter by sector, risk level, signal.	P1	MVP

5.4 Prediction & Daily Open/Close Probability

ID	Requirement	Pri	Phase
FR-PRED-01	Generate BUY/HOLD/SELL for each stock using RF + XGBoost ensemble on technical features.	P0	MVP
FR-PRED-02	Show calibrated probability, model version, training date and walk-forward backtest hit-rate next to every signal.	P0	MVP
FR-PRED-03	Show P(close > prev close), P(gap-up open) and P10/P50/P90 range for next open and close.	P1	MVP
FR-PRED-04	Scheduled daily refresh after 16:30 IST; weekly retraining; models versioned and rollback-able.	P0	MVP
FR-PRED-05	Prediction horizon selector (1-day / 5-day).	P2	Post

5.5 Explainable AI (SHAP)

ID	Requirement	Pri	Phase
FR-XAI-01	For every prediction display top-5 SHAP feature contributions as a signed bar chart (green/red).	P0	MVP
FR-XAI-02	Generate a template-based plain-language explanation from SHAP values (no LLM needed); optional LLM paraphrase.	P0	MVP
FR-XAI-03	Global feature-importance page; SHAP also used to explain risk-score components and recommendation ranking.	P1	MVP

5.6 Portfolio, Risk & What-If

ID	Requirement	Pri	Phase
FR-PORT-01	User can add/edit/delete holdings (symbol, amount or quantity × price); input validated against watchlist.	P0	MVP
FR-PORT-02	Show sector allocation, top concentration, diversification score; warn when sector > 40% (configurable).	P0	MVP
FR-RISK-01	Compute Portfolio Risk Score 0–100 from volatility, MDD, beta, sector concentration, correlation, single-stock risk, diversification credit.	P0	MVP
FR-RISK-02	Show contribution breakdown ("Why is my risk 64?").	P0	MVP
FR-WIF-01	What-if: user changes amounts/adds/removes a stock; system shows Before vs After for sector exposure and risk score.	P0	MVP
FR-WIF-02	Simulations can be saved and compared (max 10 per portfolio).	P2	Post

5.7 Asset Comparison, Backtesting & Dashboard

ID	Requirement	Pri	Phase
FR-CMP-01	Compare investing ₹X over N years in selected stock(s) vs Gold vs FD using historical CAGR and a user-adjustable FD rate; chart + table.	P0	MVP
FR-CMP-02	Clearly label results as historical illustration, not forecast; optional inflation adjustment.	P1	MVP
FR-BT-01	Backtest AI strategy vs Buy & Hold: return, max drawdown, win rate, Sharpe, with transaction cost assumption.	P1	MVP
FR-DASH-01	Dashboard: portfolio value, risk score, today's signals count, high-risk holdings, top personalised picks.	P0	MVP
FR-DASH-02	Interactive price chart with indicators overlay (Plotly).	P0	MVP

5.8 RAG Chatbot

ID	Requirement	Pri	Phase
FR-CHAT-01	Chatbot answers SEBI/investor-protection questions only from the indexed knowledge base, with source document + page citation.	P0	MVP
FR-CHAT-02	If no relevant context is retrieved (score below threshold) the bot says it cannot find it — no guessing.	P0	MVP
FR-CHAT-03	Bot refuses to give direct buy/sell instructions; redirects to dashboard signals with disclaimer.	P0	MVP
FR-CHAT-04	Conversation history saved per user; thumbs-up/down feedback on each answer.	P1	MVP
FR-CHAT-05	Prompt-injection filtering on user input and retrieved text.	P0	MVP

5.9 Admin, Notifications & Misc.

ID	Requirement	Pri	Phase
FR-ADM-01	Admin can upload/replace/delete SEBI documents, trigger re-index, see document version and status.	P0	MVP
FR-ADM-02	Admin dashboard: user count, daily active users, LLM calls/tokens/cost, errors, model version, last ETL run.	P1	MVP
FR-ADM-03	Admin can trigger ETL/retrain, toggle feature flags, disable users, view audit logs.	P1	MVP
FR-NOT-01	In-app notification banner for new signals / high-risk holding; email alerts later.	P2	Post
FR-GEN-01	Persistent disclaimer on every analytical page; user must accept terms/consent at registration.	P0	MVP
FR-GEN-02	CSV export of portfolio and analysis report as PDF.	P2	Post

6. Data Design & ER Diagram

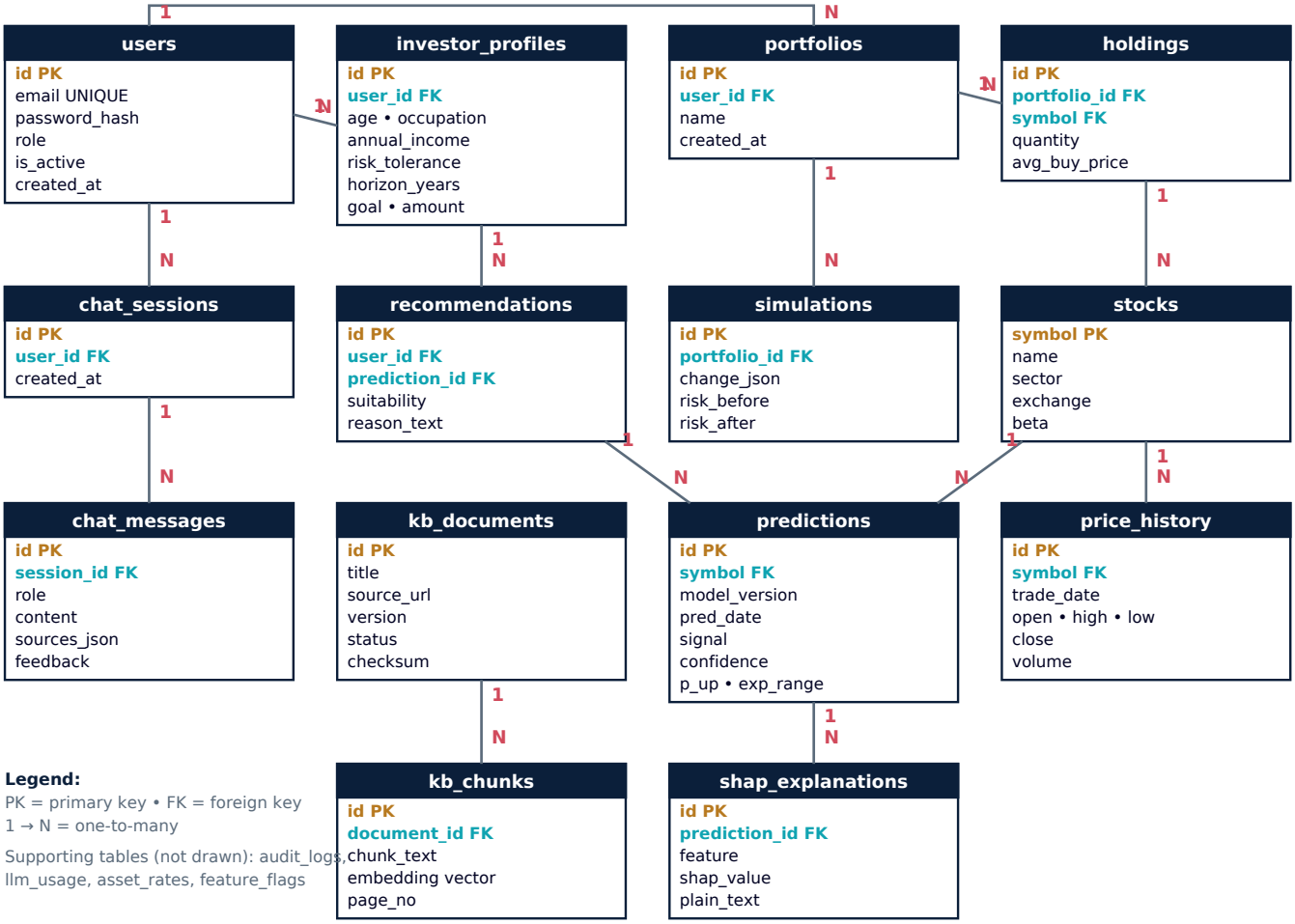


Figure 6 — Entity-Relationship diagram (core tables)

6.1 Entity Descriptions

Table	Key fields & types	Indexes / constraints
users	id UUID PK, email VARCHAR UNIQUE, password_hash TEXT, role ENUM(investor, admin), is_active BOOL, created_at TIMESTAMPTZ	UNIQUE(email)
investor_profiles	id PK, user_id FK UNIQUE, age INT, occupation VARCHAR, annual_income NUMERIC, risk_tolerance ENUM, horizon_years INT, goal VARCHAR, invest_amount NUMERIC, risk_score_computed INT	FK user_id; CHECK(age 18–100)
portfolios / holdings	portfolios(id, user_id FK, name); holdings(id, portfolio_id FK, symbol FK, quantity NUMERIC, avg_buy_price NUMERIC)	UNIQUE(portfolio_id, symbol)
stocks	symbol VARCHAR PK, name, sector, exchange, beta NUMERIC, is_active BOOL	idx(sector)
price_history	id BIGINT PK, symbol FK, trade_date DATE, open/high/low/close NUMERIC, volume BIGINT	UNIQUE(symbol, trade_date)
predictions	id PK, symbol FK, pred_date DATE, model_version, signal ENUM, confidence NUMERIC, p_up, p_gap_up, open_p10/50/90, close_p10/50/90	UNIQUE(symbol, pred_date, model_version)
shap_explanations	id PK, prediction_id FK, feature VARCHAR, shap_value NUMERIC, rank INT, plain_text TEXT	idx(prediction_id)
recommendations	id PK, user_id FK, prediction_id FK, suitability ENUM, reason_text, created_at	idx(user_id, created_at)
simulations	id PK, portfolio_id FK, change_json JSONB, exposure_before/after JSONB, risk_before INT, risk_after INT, created_at	idx(portfolio_id)
chat_sessions / chat_messages	sessions(id, user_id FK, created_at); messages(id, session_id FK, role, content, sources_json JSONB, feedback SMALLINT, latency_ms, prompt_version)	idx(session_id, created_at)
kb_documents / kb_chunks	documents(id, title, source_url, version, status ENUM, checksum, uploaded_by); chunks(id, document_id FK, chunk_text, embedding VECTOR(384/1024), page_no, tsv TSVECTOR)	HNSW idx(embedding); GIN idx(tsv); UNIQUE(checksum)

Table	Key fields & types	Indexes / constraints
audit_logs, llm_usage, asset_rates, feature_flags	audit(user_id, action, ip, ts); llm_usage(user_id, model, in/out tokens, cost, ts); asset_rates(asset_type, date, value); flags(key, enabled)	idx(ts), idx(user_id)

7. API Inventory

Base path **/api/v1**. All endpoints except register/login/health require **Bearer JWT**. Admin endpoints require role = admin.

API	Type	Method	Purpose	Auth	MVP
/auth/register • /auth/login • /auth/refresh	Internal	POST	Account creation, tokens	Public	Yes
/users/me • /users/me/export • /users/me	Internal	GET/DELETE	Profile data, export, deletion	JWT	Yes
/profile	Internal	GET/PUT	Investor profile + computed risk band	JWT	Yes
/stocks • /stocks/{sym}/prices	Internal	GET	Watchlist, OHLCV + indicators	JWT	Yes
/predictions/{sym}	Internal	GET	Signal, probability, open/close ranges	JWT	Yes
/predictions/{sym}/explain	Internal	GET	SHAP values + plain text	JWT	Yes
/recommendations	Internal	GET	Personalised ranked picks + suitability	JWT	Yes
/portfolios • /portfolios/{id}/holdings	Internal	CRUD	Manage portfolio and holdings	JWT	Yes
/portfolios/{id}/analysis	Internal	GET	Sector mix, risk score, breakdown	JWT	Yes
/portfolios/{id}/simulate	Internal	POST	What-if before/after	JWT	Yes
/compare/assets	Internal	POST	Stock vs Gold vs FD	JWT	Yes
/backtest/{sym}	Internal	GET	Strategy vs Buy & Hold metrics	JWT	Yes
/chat • /chat/sessions • /chat/{id}/feedback	Internal	POST/GET	RAG Q&A, history, feedback	JWT	Yes
/dashboard	Internal	GET	Aggregated dashboard payload	JWT	Yes
/admin/documents • /admin/reindex • /admin/jobs/etl • /admin/jobs/retrain • /admin/usage • /admin/logs	Internal	POST/GET	KB management, jobs, monitoring	Admin	Yes
/health • /ready	Internal	GET	Liveness / readiness	Public	Yes
yfinance / NSE bhavcopy	External	Library / HTTP	Daily OHLCV, metadata (retry ×3, exp. backoff, cache, fallback)	None / key	Yes
LLM provider API (Gemini / Claude)	External	POST	Chat completion (timeout 20s, retry ×2, fallback to next provider/local)	API key	Yes
News API / RSS	External	GET	Headlines for sentiment	Key	Post
Email (Brevo/Resend)	External	POST	Verification, alerts	Key	Post

8. AI / ML / RAG Design

8.1 Prediction Model Specification

Item	Specification
Target	3-class label from next-day (or 5-day) return: BUY if > +0.5%, SELL if < −0.5%, else HOLD (threshold tunable). Binary up/down head used for P(up).
Features	Returns (1/5/10d), SMA20/50, EMA12/26, MACD + signal, RSI14, Bollinger %B & width, ATR, rolling volatility (10/20d), volume ratio, gap %, day-of-week; optional Nifty-50 index return as market regime.
Models	Random Forest + XGBoost ensemble (soft voting); Logistic Regression as baseline. Quantile XGBoost for open/close P10/P50/P90.
Validation	Walk-forward (expanding window), 5+ folds; no shuffling; purge gap equal to horizon; hyper-parameter search via Optuna/RandomizedSearch inside folds.
Calibration	Isotonic / Platt calibration; report Brier score and reliability curve.
Metrics	Accuracy, macro-F1, precision/recall per class, AUC, Brier; strategy: CAGR, Sharpe, MDD, win rate vs Buy & Hold. Release gate: must beat baseline on out-of-sample folds.
Retraining	Weekly; compare to champion model; auto-promote only if metrics not worse; keep last 3 versions.

8.2 Personalization & Risk Formulas (deterministic)

Component	Rule
Risk capacity (0–100)	40% horizon (≥5y = 100, 3y = 60, <1y = 10) + 30% income/investment ratio + 20% age band + 10% occupation stability (salaried/ business/ freelancer/ retired).
Risk willingness	5-question questionnaire → 0–100. Final band = min(capacity band, willingness band) to be conservative.
Suitability	Compare stock 1-yr annual volatility & beta with band limits (Low ≤ 22% vol, Moderate ≤ 32%, High > 32%; configurable). Within band = HIGH, one band above = MODERATE, two bands = LOW.
Portfolio Risk Score	Weighted 0–100: sector concentration 25 + volatility 25 + correlation 15 + max drawdown 15 + beta 10 + single-stock concentration 10 − diversification credit (up to 10). Bands: 0–30 Low, 31–55 Moderate, 56–75 Moderate-High, 76–100 High.
Gold / FD comparison	Stock & Gold: historical CAGR from price series. FD: compound interest with user/admin-set rate. Output: final value, CAGR, worst drawdown (stock/gold), real return after inflation (optional).

8.3 Where LLMs are used — and where they are not

AI Feature	Approach	LLM?	MVP
Stock signal & probabilities	ML ensemble (XGBoost/RF)	No	Yes
SHAP explanation text	Template sentences from top SHAP features (deterministic, auditable)	No (optional polish)	Yes
Personal suitability, risk score, what-if, Gold/FD comparison	Pure Python / pandas	No	Yes
SEBI Q&A chatbot	RAG + LLM with citations	Yes	Yes
Natural-language summary of portfolio report	LLM given structured JSON of computed results (cannot alter numbers)	Yes	Post
News sentiment	FinBERT classifier (not a generative LLM)	No	Post

8.4 Model, Embedding & Vector Store Selection

Need	Choice (Primary)	Backup / Alternative	Reason
Chat LLM	Gemini Flash-class (free tier in dev)	Claude Haiku-class; Ollama Llama 3 / Qwen (local)	Cheap, fast, large context; router prevents lock-in. Verify current limits/pricing.
Embeddings	Dev: bge-small-en-v1.5 (384-d) • MVP: bge-m3 / bge-base (768–1024-d)	OpenAI text-embedding-3-small, Google embeddings	Free, self-hosted, strong retrieval, no data sent out
Reranker	bge-reranker-base	Cohere Rerank (paid)	Improves top-5 precision for small corpus
Vector DB	PostgreSQL + pgvector (HNSW)	Qdrant / Chroma / FAISS	Corpus is small (< 50k chunks); one database to operate; hybrid search with tsvector; transactional deletes

8.5 RAG Design Details

Aspect	Decision
Knowledge sources	SEBI circulars, investor-charter, SEBI (Investment Advisers) & (Research Analysts) regulations, mutual-fund/stock-market investor FAQs, platform FAQ (Markdown). Downloaded from sebi.gov.in; no scraping of paywalled sources.
Chunking	Structure-aware: split by headings/clauses, 500–800 tokens, 80-token overlap; keep clause numbers intact.
Metadata	doc_id, title, circular date, version, section, page, source URL, status (active/superseded).
Retrieval	Hybrid: vector top-20 + Postgres full-text top-20 → Reciprocal Rank Fusion → rerank → top-5; similarity threshold for “no answer”.
Prompting	System rules: answer only from context, cite [doc, page], say “not found” otherwise, no buy/sell advice, temperature 0–0.2, structured JSON output (answer, citations, confidence).
Hallucination control	Citation validation (cited chunk IDs must exist in retrieved set); refusal on low retrieval score; faithfulness check in evaluation.
Versioning / deletion	Checksum de-duplication; new version marks old chunks superseded; delete cascades to chunks and embeddings; full re-index command.
Access control	KB is public-regulatory content; chat history strictly per user; admin-only document writes.

8.6 RAG & Model Evaluation

Metric	Target (MVP)	How measured
Retrieval precision@5 / recall@5	≥ 0.70 / ≥ 0.85	50-question golden set with labelled source chunks

Metric	Target (MVP)	How measured
Faithfulness / answer correctness	≥ 0.85 / ≥ 0.80	RAGAS + manual review of 30 answers
Citation correctness	≥ 95%	Automated check: cited text supports claim
Hallucination rate	≤ 5%; 100% refusal on 10 out-of-scope questions	Adversarial set incl. “which stock will double?”
P95 chat latency	≤ 8 s	Request logs
Prediction model	Beats baseline on walk-forward OOS; Brier ≤ 0.25	Backtest report generated per training run

9. Non-Functional Requirements

Category	MVP Target
Performance	Dashboard/API responses ≤ 2 s (P95) from precomputed signals; what-if ≤ 3 s; chatbot ≤ 8 s; daily batch for 50 stocks ≤ 15 min.
Scalability	100 concurrent users on a single 2 vCPU/4 GB node; stateless API enables horizontal scale; DB read-replica later.
Availability / Reliability	≥ 99% monthly (demo/MVP); graceful degradation: cached signals if data API fails, local LLM if provider fails.
Security	OWASP Top-10 mitigations; HTTPS only; no critical/high Bandit or pip-audit findings at release.
Maintainability	Modular code, type hints, ≥ 70% test coverage on core logic, Ruff/Black enforced in CI, OpenAPI docs.
Observability	Structured JSON logs, request IDs, Sentry errors, /health + uptime monitor, LLM token/cost dashboard.
Usability / Accessibility	Responsive layout; colour + icon (not colour alone) for signals; plain-language explanations; WCAG 2.1 AA aims.
Localization	English (MVP); ₹ currency and Indian number format; IST timestamps.
Backup / Recovery	Daily automated DB backup (7-day retention); RPO 24 h; RTO 4 h; restore tested once before launch.
Data retention	Prices & predictions: kept; chat history: 12 months; audit logs: 12 months; deleted user data purged within 30 days.

10. Security, Privacy & Compliance

Area	MVP requirement	Post-MVP
AuthN / Session	bcrypt/argon2, JWT 15–30 min + refresh rotation, HTTPS, login throttling	MFA, email verification, device sessions
AuthZ / RBAC	Role checks in dependency layer; object-level ownership checks (user can only read own portfolios/chats)	Fine-grained permissions
API security	Pydantic validation, rate limiting (slowapi), CORS allow-list, secure headers, request size limits	WAF, API gateway
Injection / XSS / CSRF	ORM parameterised queries only; output escaping; no raw HTML from LLM; token in header (not cookie) or SameSite cookies + CSRF token	SAST/DAST in CI
Secrets	.env locally, platform secret manager in deployment; never committed; rotate keys	Vault / cloud KMS
LLM-specific	Prompt-injection filter, context delimiting, no tools with side effects, output validation, never send passwords/PII to LLM	Red-team test suite, LLM firewall
RAG poisoning	Only admin can ingest; checksum + source whitelist; review before activation	Document signing
File upload (admin)	PDF only, size ≤ 25 MB, MIME check, virus scan optional, stored outside web root	Malware scanning
Audit	Log login, role changes, doc upload/delete, admin actions	Tamper-evident logs

10.1 Privacy & Data Protection (India — DPDP Act 2023)

Data	Purpose	Stored	Shared with AI provider?
Email, password hash	Account	PostgreSQL (hashed)	No
Age, occupation, income, risk, goal	Personalisation	PostgreSQL; minimal fields only	No — personalisation is computed locally
Portfolio holdings	Risk analysis	PostgreSQL	No (LLM receives only anonymised aggregates if summary feature is enabled)

Data	Purpose	Stored	Shared with AI provider?
Chat questions	Q&A, quality improvement	PostgreSQL, 12 months	Yes — question text + retrieved SEBI context only; avoid PII, warn users

- Explicit consent checkbox at registration (purpose-specific), privacy notice, right to access/export/correct/erase, grievance contact email.
- Prefer self-hosted embeddings and optional local LLM so sensitive financial data never leaves the server.
- **Regulatory flag:** Personalised buy/sell recommendations to the public may fall under SEBI (Investment Advisers) Regulations, 2013 and research-analyst rules. For production/commercial use, obtain legal review or registration; keep academic version clearly labelled.

11. Testing & Acceptance Criteria

Module	Test types	Critical scenarios	Pri
Auth	Unit, API, security	Duplicate email; wrong password ×5; expired/tampered JWT; admin route as investor	P0
Data/ETL	Unit, integration	API timeout → fallback; missing days; split adjustment; duplicate insert idempotent	P0
ML pipeline	Unit, model validation	No look-ahead leakage test; reproducible with seed; beats baseline; calibration	P0
SHAP / explanation	Unit	Values sum to (output − base); top-5 ordering; text generated for each signal	P1
Risk / what-if	Unit (golden values)	Single stock=100%; empty portfolio; weights sum 100%; before/after consistency	P0
RAG chatbot	RAG eval, adversarial	Out-of-scope query refusal; prompt injection; wrong/no retrieval; citation validity	P0
UI	Manual + Playwright smoke	Login → profile → signal → portfolio → what-if on desktop & mobile width	P1
Performance	Locust	100 concurrent users on dashboard and chat	P1
Security	Bandit, pip-audit, OWASP ZAP baseline	SQLi/XSS payloads; IDOR on portfolio IDs; rate-limit	P0

11.1 Acceptance Criteria (Given / When / Then)

ID	Criterion
AC-01 Login	**Given** a registered user, **when** valid credentials are entered, **then** a JWT is issued and the user lands on the dashboard; invalid credentials show a generic error and never reveal which field was wrong.
AC-02 Profile	**Given** a logged-in user, **when** the profile form is saved with valid data, **then** a Risk Band is computed and shown; invalid age/amount is rejected with a field-level message.
AC-03 Signal	**Given** a watchlist stock, **when** opened, **then** signal, calibrated probability, model version, data date and backtest hit-rate are displayed with the disclaimer.
AC-04 SHAP	**Given** a prediction, **when** “Why?” is opened, **then** top-5 signed contributions and a plain-language paragraph are shown.
AC-05 Suitability	**Given** a Low-risk user and a high-volatility BUY stock, **then** suitability is LOW/MODERATE with a reason referencing volatility.
AC-06 Portfolio	**Given** holdings totalling ₹50,000, **then** sector percentages sum to 100%, risk score 0–100 with breakdown is shown, and a warning appears if one sector > 40%.
AC-07 What-if	**Given** a saved portfolio, **when** TCS is raised from ₹15,000 to ₹25,000, **then** IT exposure and risk score recompute and the original portfolio is unchanged.
AC-08 Compare	**Given** ₹50,000 and 5 years, **then** stock, Gold and FD values are shown with CAGR and a “historical, not a forecast” label.
AC-09 Chatbot	**Given** a SEBI question in the KB, **then** the answer includes ≥ 1 valid citation; **given** an out-of-scope or “which stock to buy” question, **then** the bot declines and refers to the dashboard.
AC-10 Admin	**Given** an admin, **when** a PDF is uploaded, **then** it is parsed, chunked, embedded and searchable within 5 minutes; deleting it removes its chunks.

12. Deployment, DevOps & Cost

12.1 Environments & Hosting

Environment	Needed?	Setup
Local	Yes	Docker Compose: api + streamlit + postgres(pgvector); SQLite/Ollama optional
Staging / Demo	Yes (one)	Render / Railway / HF Spaces free or low-cost tier; used for QA + UAT with mentor
Production	Yes (when public)	AWS Lightsail / EC2 (ap-south-1 Mumbai) or Azure/GCP India region; managed Postgres optional
Separate Dev & QA	No	Merged into Local + Staging to avoid overhead

Platform	Free tier	Ease	Production fit	Verdict
Render	Free web service (sleeps), small paid plans	High	Good	Recommended demo host
Railway	Trial credits / low usage-based	High	Good	Alternative demo host
Hugging Face Spaces	Free CPU Space	High	Demo only	Good for Streamlit showcase
AWS (Lightsail/EC2)	12-month free tier (account dependent)	Medium	Excellent, Mumbai region	Recommended production
Azure / GCP	Student credits / free tier	Medium	Excellent	Alternative
Supabase	Free Postgres + pgvector	High	Good	Optional managed DB
Vercel / Cloudflare	Generous free	High	Frontend only	Needed only if React frontend is adopted

12.2 DevOps Practice (MVP-sized)

- **Git:** GitHub; **main** (protected) + **develop** + short-lived **feature/*** branches; conventional commits; PR review by the other member.
- **CI (GitHub Actions):** Ruff/Black → pytest + coverage → Bandit + pip-audit → Docker build. **CD:** auto-deploy develop → staging; manual approval main → production.
- **Migrations:** Alembic; backward-compatible changes; rollback = previous Docker image + *alembic downgrade*.
- **Ops:** /health checks, UptimeRobot, Sentry, JSON logs, nightly *pg_dump* to object storage, quarterly restore drill.

12.3 Estimated Monthly Cost (indicative — verify current official pricing)

Component	Dev / Free	Small MVP	1K users	10K users
Hosting (API + UI)	₹0 (local / free tier)	₹0–800	₹1,500–4,000	₹10,000–25,000
PostgreSQL + pgvector	₹0 (Docker)	₹0–500	₹1,000–2,500	₹5,000–15,000
Market data	₹0 (yfinance)	₹0	₹0–2,000	₹4,000–10,000 (licensed feed)
LLM API (chat only)	₹0 (free tier / Ollama)	₹0–300	₹500–2,500	₹5,000–25,000
Embeddings / reranker	₹0 (self-hosted)	₹0	₹0	₹0–3,000 (GPU/CPU upsizing)
Email / monitoring	₹0	₹0	₹0–1,000	₹2,000–5,000
Total (approx.)	**₹0**	**₹0–1,600**	**₹3,000–12,000**	**₹26,000–83,000**

Assumptions: 20 chat requests/user/month, ~1,500 tokens/request, 1 portfolio/user, predictions precomputed daily (no per-user ML cost). Free tiers change; "free signup" ≠ free production.

13. Risk Register

Risk	Prob.	Impact	Mitigation	Owner
Model accuracy near chance; users over-trust signals	High	High	Calibration, show backtest, disclaimers, HOLD zone, avoid "guaranteed" language	ML lead
Data leakage inflating accuracy	Med	High	Walk-forward CV, leakage unit tests, mentor review	ML lead
yfinance breaks / rate-limits / licensing	High	Med	Adapter layer, caching, NSE bhavcopy fallback, store history locally	Backend
SEBI regulatory exposure	Med	High	Educational positioning, disclaimers, legal review before public launch	PM
LLM hallucination / prompt injection	Med	High	Grounded RAG, citation validation, refusal, filters, eval gate	RAG

Risk	Prob.	Impact	Mitigation	Owner
LLM free-tier limits/cost spike	Med	Med	Router + local Ollama fallback, caching, per-user quotas	LLM
Security breach / PII leak	Low	High	Hashing, RBAC, HTTPS, minimal PII, secret management	Backend
Scope creep (sentiment, mobile, alerts)	High	Med	Strict MVP boundary in §1.2; backlog for Post-MVP	PM
Free-hosting cold starts / memory limits (SHAP, embeddings)	Med	Med	Precompute SHAP daily, small embedding model, lazy loading	DevOps
Two-person team bandwidth	High	Med	Phase gating, parallel tracks, mentor check-ins weekly	PM

14. Delivery Plan, Team & Definition of Done

14.1 Development Phases

Phase	Objective	Deliverables / Exit criteria
0 — Discovery	Freeze scope, watchlist, SEBI doc list	Approved SRS, repo, Docker skeleton, CI green
1 — Foundation	DB, auth, profile, data ETL	Migrations, register/login, profile form, 5 yrs OHLCV loaded
2 — ML Core	Features, models, calibration, backtest, SHAP	Model report beating baseline; /predictions + /explain working
3 — Portfolio & Personalization	Risk engine, suitability, what-if, Gold/FD comparison	Golden-value unit tests pass; AC-05…AC-08 met
4 — RAG	Ingestion, hybrid retrieval, chatbot, admin upload	Eval targets in §8.6 met; AC-09/10 met
5 — Dashboard & Polish	Streamlit pages, charts, disclaimers, admin panel	All pages integrated; UX review
6 — Testing & Hardening	Security, performance, UAT	Minimum test suite green; no high-severity bugs
7 — Deployment	Staging → production, monitoring, backup	Live URL, health checks, restore tested, user guide + runbook

14.2 Development Order & Parallel Tracks

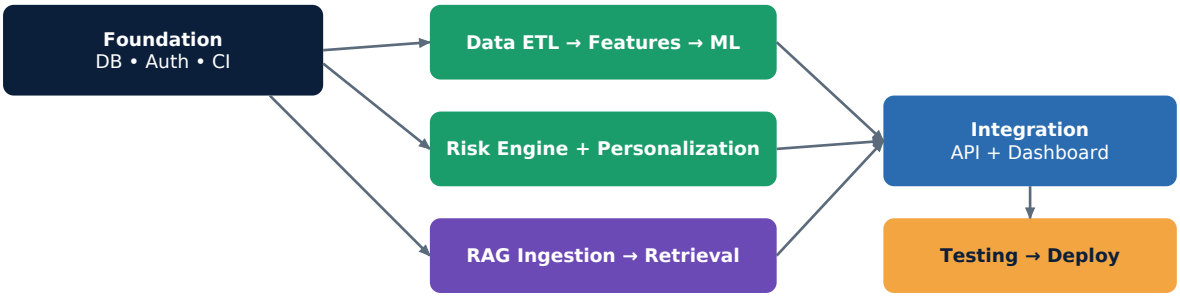


Figure 7 — Three parallel tracks (ML, Risk, RAG) converge at integration

14.3 Team Responsibility (Roles are shared in a 2-person team)

Module	Sujeet	Anuj	Mentor (Samiksha More)
Backend, Auth, DB, DevOps, deployment	Lead	Support	Review
Data ETL, ML models, SHAP, backtesting	Support	Lead	Validate methodology
Risk engine, personalization, what-if, Gold/FD	Lead	Support	Review formulas
RAG, LLM router, prompts, evaluation	Support	Lead	Review
Streamlit UI / dashboards	Co-own	Co-own	UX feedback
QA, security checks, documentation	Co-own	Co-own	Acceptance (UAT)

14.4 Definition of Done

- Code complete, peer-reviewed via PR, Ruff/Black clean; type hints on public functions.
- Unit + integration tests written and passing; coverage ≥ 70% on core modules; API documented in OpenAPI.
- Security checks pass (Bandit, pip-audit); no secrets in repo; env variables documented in *.env.example*.
- ML: walk-forward report stored; RAG: evaluation targets met; prompts versioned.
- Error handling and structured logging present; edge cases (empty input, API timeout, no retrieval) covered.

- Deployed to staging, QA-verified, accepted by mentor in UAT; runbook and user guide updated.

14.5 Decisions Required (blocking first)

Decision	Recommended	Alternative	Impact if changed
Frontend	Streamlit	React + FastAPI	+3–4 weeks
Watchlist size	30–50 NSE stocks	Nifty 500	Training/ETL time ×10
Prediction horizon	Next-day + 5-day	Intraday	Needs paid tick data
LLM provider	Gemini free tier + Ollama fallback	Claude / OpenAI paid	Small cost; router makes swap easy
Positioning	Educational decision-support	Registered advisory	Legal/registration effort

Appendix A. Mandatory UI Disclaimer

Display on every analytical page and at registration

• StockIQ is an educational decision-support tool. Outputs are model-based estimates, not investment advice, recommendations or guarantees of return.

• Past performance and backtests do not indicate future results. Markets are subject to risk, including loss of capital.

• StockIQ is not registered with SEBI as an Investment Adviser or Research Analyst. Consult a SEBI-registered adviser before investing.